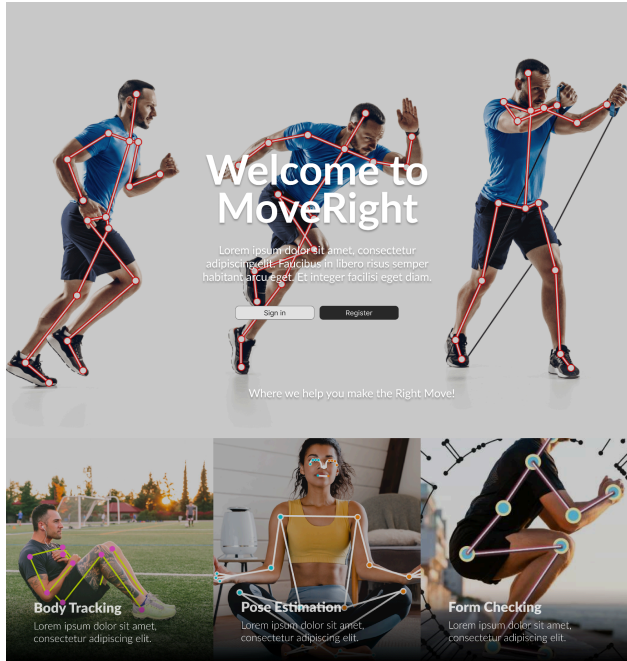




This web application is a machine learning and math based movement solution designed to change the way of the personal trainer. The application is designed to simplify exercising feedback by using the consumer's camera for live recording. Feedback trainers and models in the form of human silhouettes track the user as they exercise and return to the user simple form cues to fix their training and get the best out of exercising.

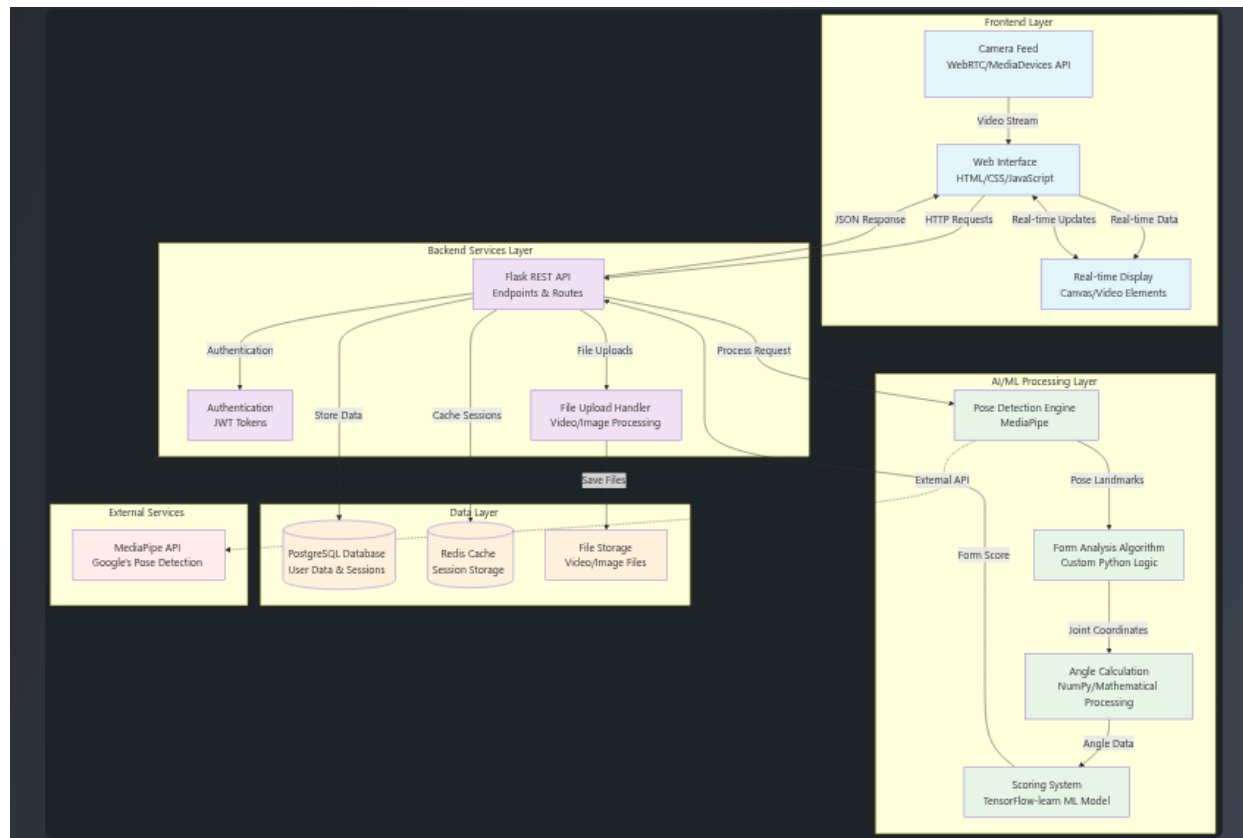
User and stories

Our web application is targeted to an audience who wish to exercise independently and are looking for a portable alternative to a trainer. As such, the application was designed for the task of giving form cues to improve their exercise form by capturing a set of various repetitions of a given exercise and using machine learning to pinpoint areas of form breakdown. This information is provided as simple commands for the user to follow as they finish their set.

Our customers can readily interact with this model as soon as an account is created, with the opportunity to test the technology on a variety of three bodyweight exercises: pushups, squats and crunches. Only a single recording session for each exercise, up to a total amount of three feedback sessions daily, will be allowed every day. Should the consumer desire to store their recorded session to either view their progress and track it, Move Right offers viewers to store each of their daily recording sessions in a persistent database that is tied to their account.

Tech stack

The application works with the following packages and dependencies:

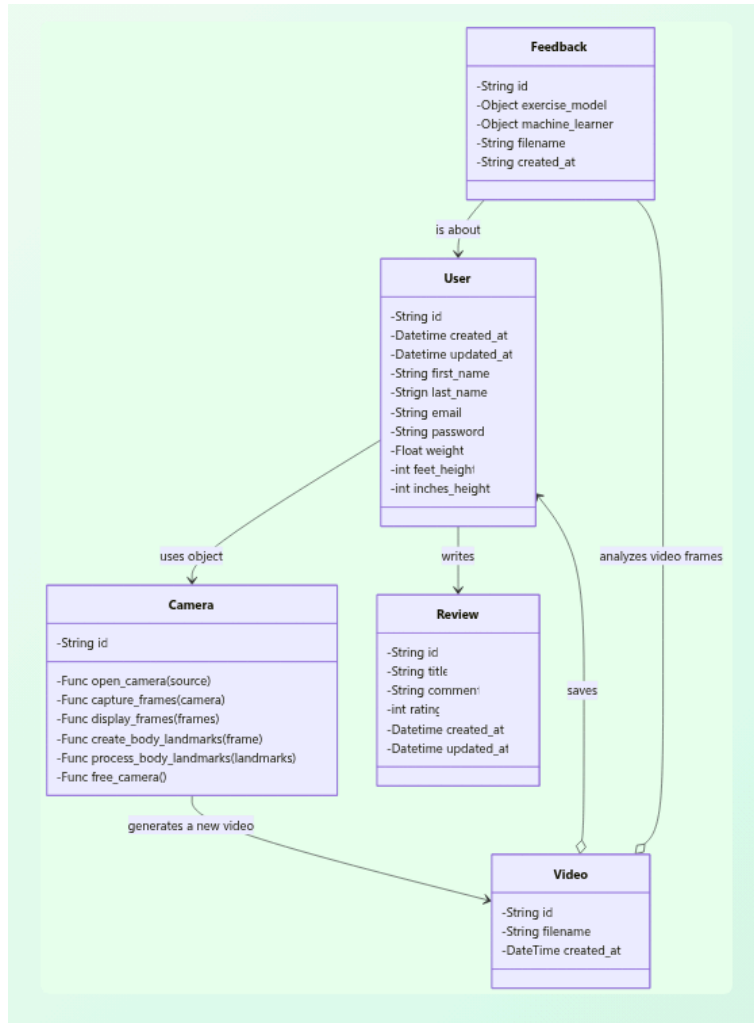


Entities and relationships:

To address interface interactions and API endpoints, the following entities and their relationships have been established:

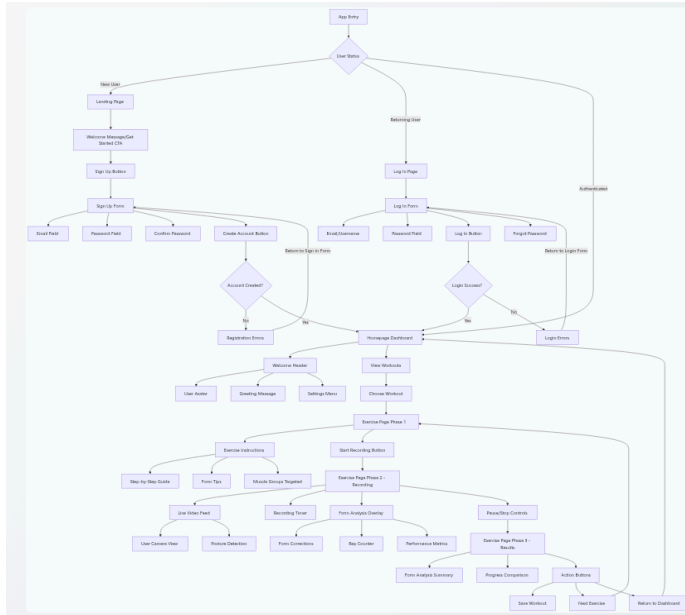
1. **User:** refers to the user creating an account. The main focus
2. **Camera:** the interactable object of the application, it works by retrieving source code for image capture and processing.
3. **Video:** a result object from the image captured by the camera, these are items that can be stored in a database and are classified by the user's name and the date at which it was recorded.

4. Feedback: the machine learning model that is loaded into the camera object for image processing and data analysis to generate correction cues and prompt the user to correct their form.
5. Review: stores comments and feedback the consumer might have about MOve Right. Only one per user is allowed.

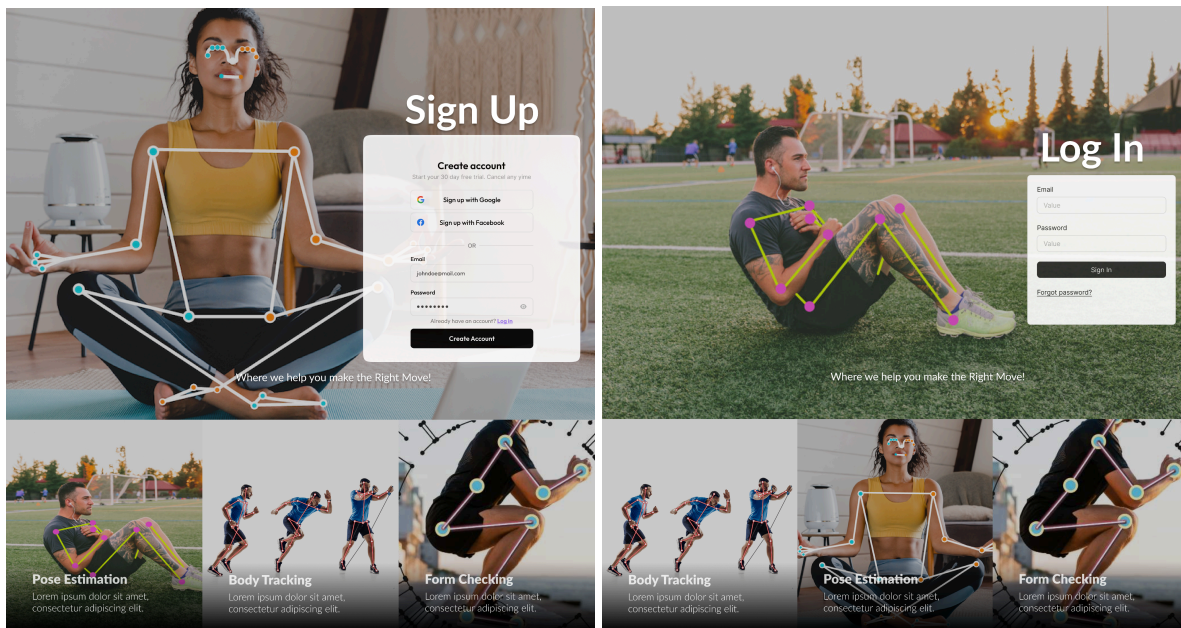


Interface and sequences:

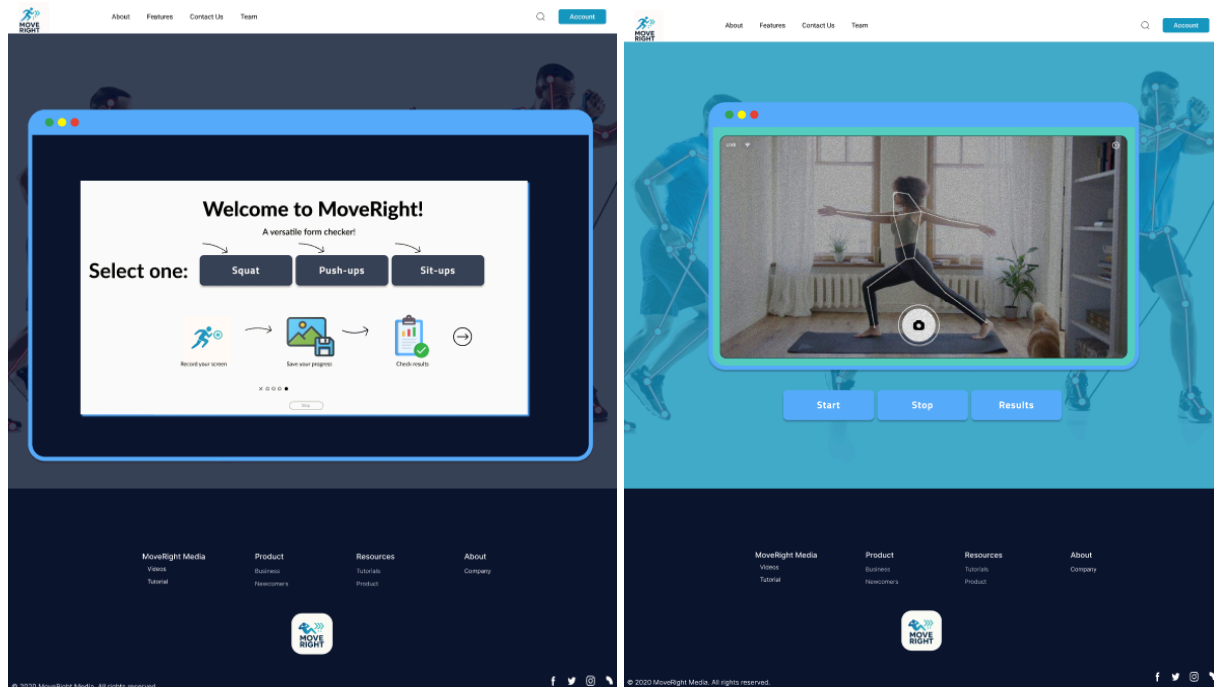
Move Right was developed with simplicity in mind, giving customers a very usable service and easy interaction with each of its components. The following is a render of all interactions performed by the user:



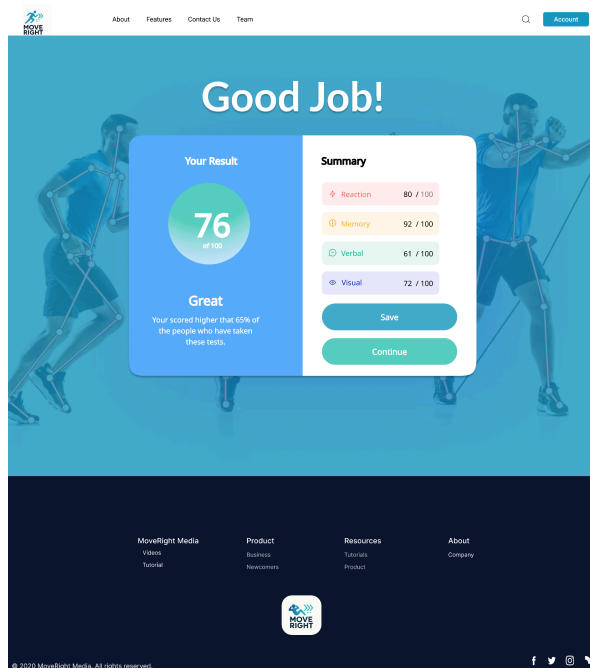
The User will be allowed to create a single profile for their unique email. The following credentials will be stored per user:
Full name, email, password, weight, and height(feet and inches).



When a user logs in, they are directed to the homepage where they will see the features, their profile and the window to start a new recording session for feedback. A new session starts by selecting the exercise they wish to train and have corrected by the feedback model:



As a session finishes, the result window shows the recorded video, the areas to correct and the overall results of their daily feedback session. The user is then given the choice of storing these recordings as they wish:



DevOps and Confluence

1. SCM and QA:

All communications are done via the following tools:

- a. Trello: agenda and goal setting
- b. Discord: virtual meetspace
- c. Whatsapp: direct communication

All testing and versions are controlled and monitored through:

- d. GitHub: version control
- e. TDD

2. Technical Justifications:

Packages were chosen for the following reasons:

Python: back end language that is familiar to all developers and stakeholders, great resources and open source tools available and solid ML and AI concepts for the project.

- a. OpenCV: the leading open source computer vision with extensive documentation, multi platform and multi paradigm project approach.
- b. MediaPipe Solutions: an easy machine learning friendly framework for image and movement pipeline processing forms. Lightweight and well explored, it is amongst the best for pose estimation.
- c. TensorFlow: the go to framework for machine learning in Python, neural networks and pipeline processing, works well with other miniature ML models, ideal for the current IDE and hardware, ideal for implementing into other languages(C++ or Java).
- d. NumPy(numerical computations)
- e. Pandas(spreadsheet creation and data structuring).

React: the ideal front end framework for Web APIs that sits within the scope of learning for project MVP. Allows porting to Mobile apps with React Native.

External APIs:

Flask: lightweight and simple, it provides all the necessary back end API endpoints and all necessary components for networking, such as migrations, CORS, web tokens with JWT and cookies, etc.

Databases:

PostgreSQL: powerful database with familiar queries and solid table schemas.

MongoDB: non relational database utilized for document and key-value paired persistence of non credential objects, such as videos and images.